DATA CENTER SIMULATION PERFORMANCE REPORT

Profile: NORMAL_DAY_SAMPLE | Generated: 2026-08-23 16:09:44

Parameter	Value
Experiment Duration	60.0 seconds
Start Time (T1)	1787481523.1421537
End Time (T2)	1787481583.1421552
Primary Bottleneck	Application / Endpoint Latency
Confidence Level	LOW_SAMPLE_SIZE
Failure Classification	SLA_NON_COMPLIANCE

SLA Metric	Measured Value	Target Limit	Result
P95 Response Time	1900.0 ms	500.0 ms	FAIL
Error Rate	0.0 %	1.0 %	PASS
Peak CPU Utilization	0.54 %	80.0 %	PASS
Peak Memory Utilization	16.77 %	90.0 %	PASS

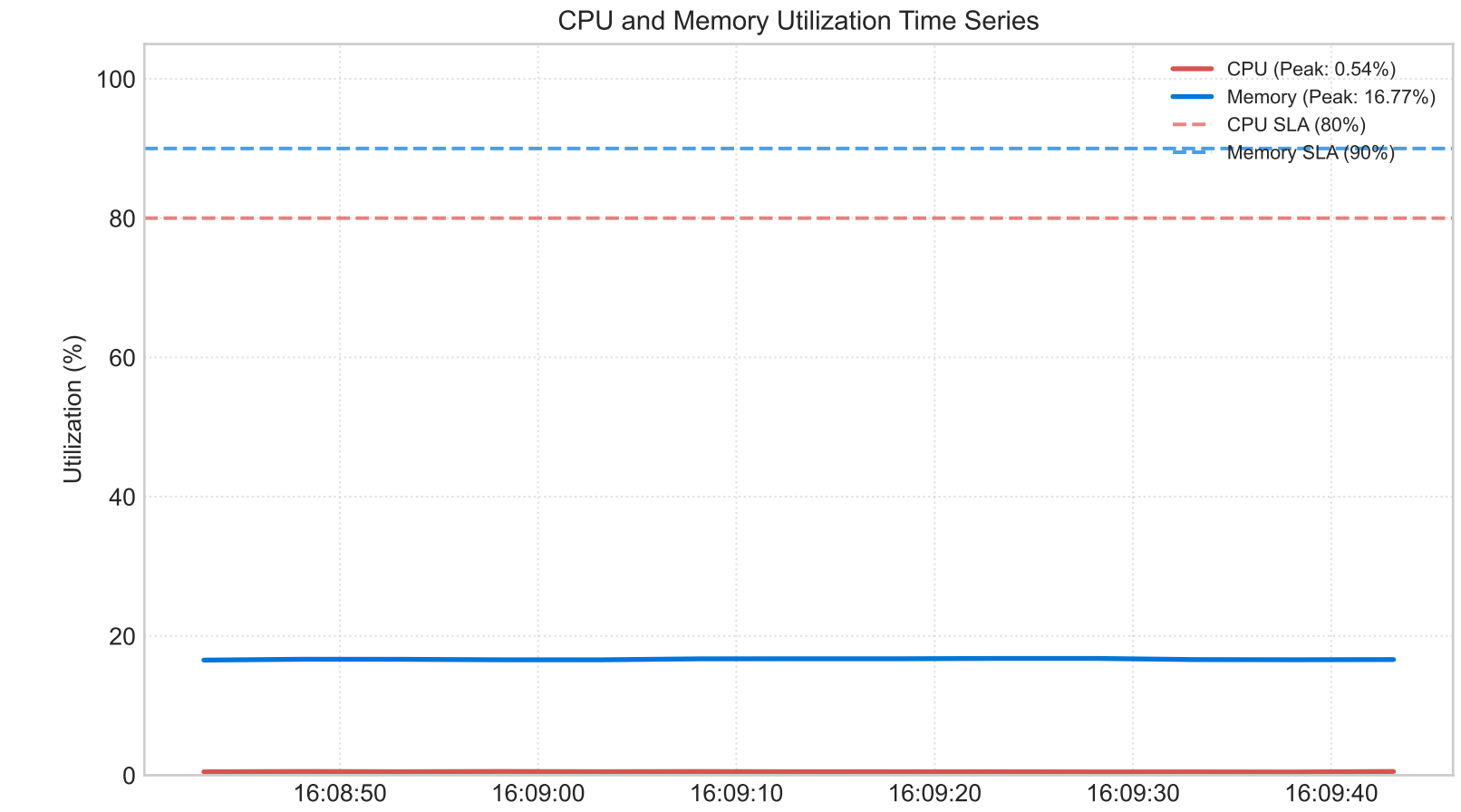
Executive Conclusion & Evidence Synthesis:

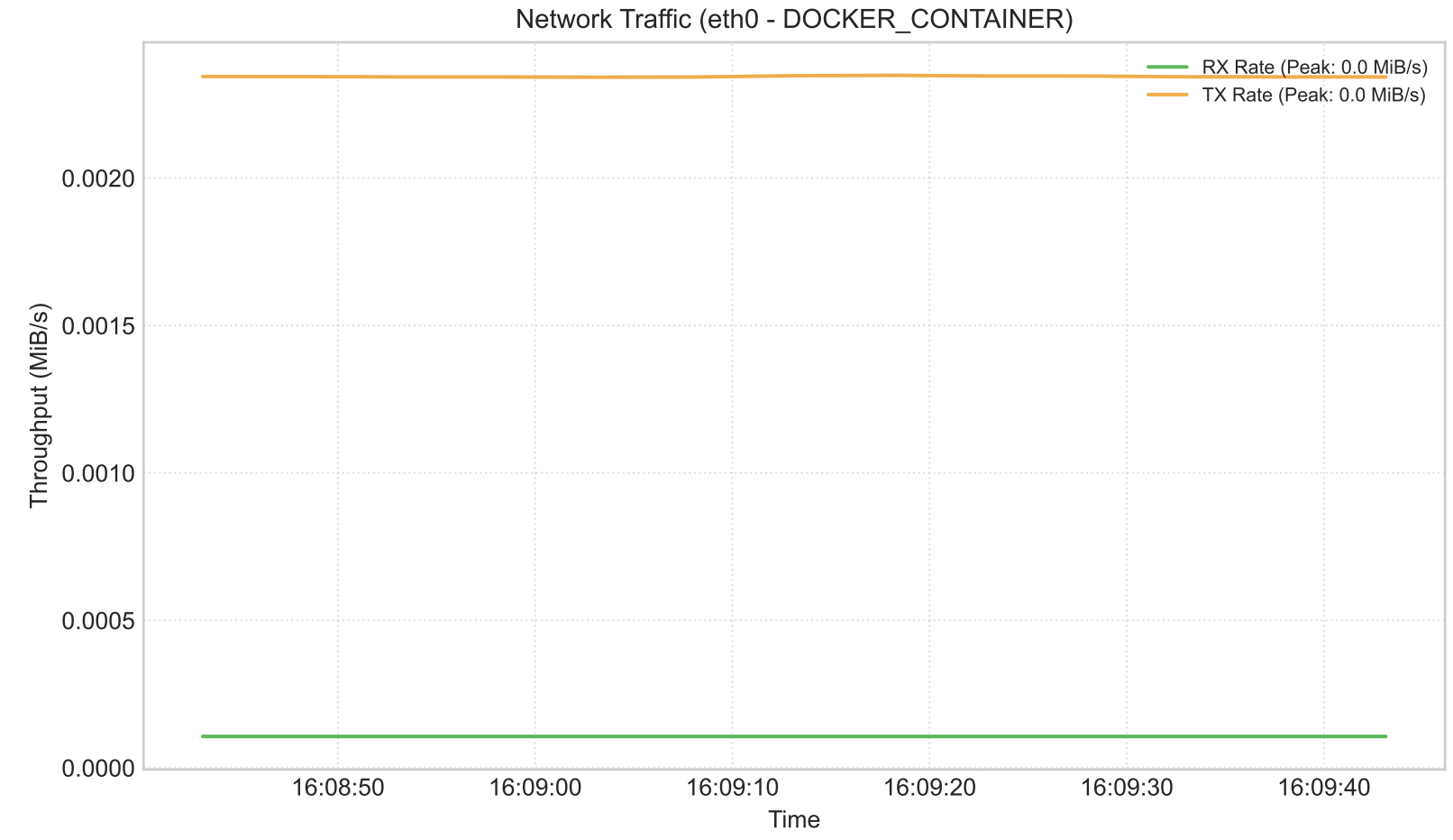
The experiment executed 41 requests with overall SLA status 'FAIL' (3/4 checks satisfied). No container crashes or service interruptions were detected by the watchdog. Primary bottleneck assessed as 'Application / Endpoint Latency' (Confidence: LOW_SAMPLE_SIZE).

Diagnostic Observations:

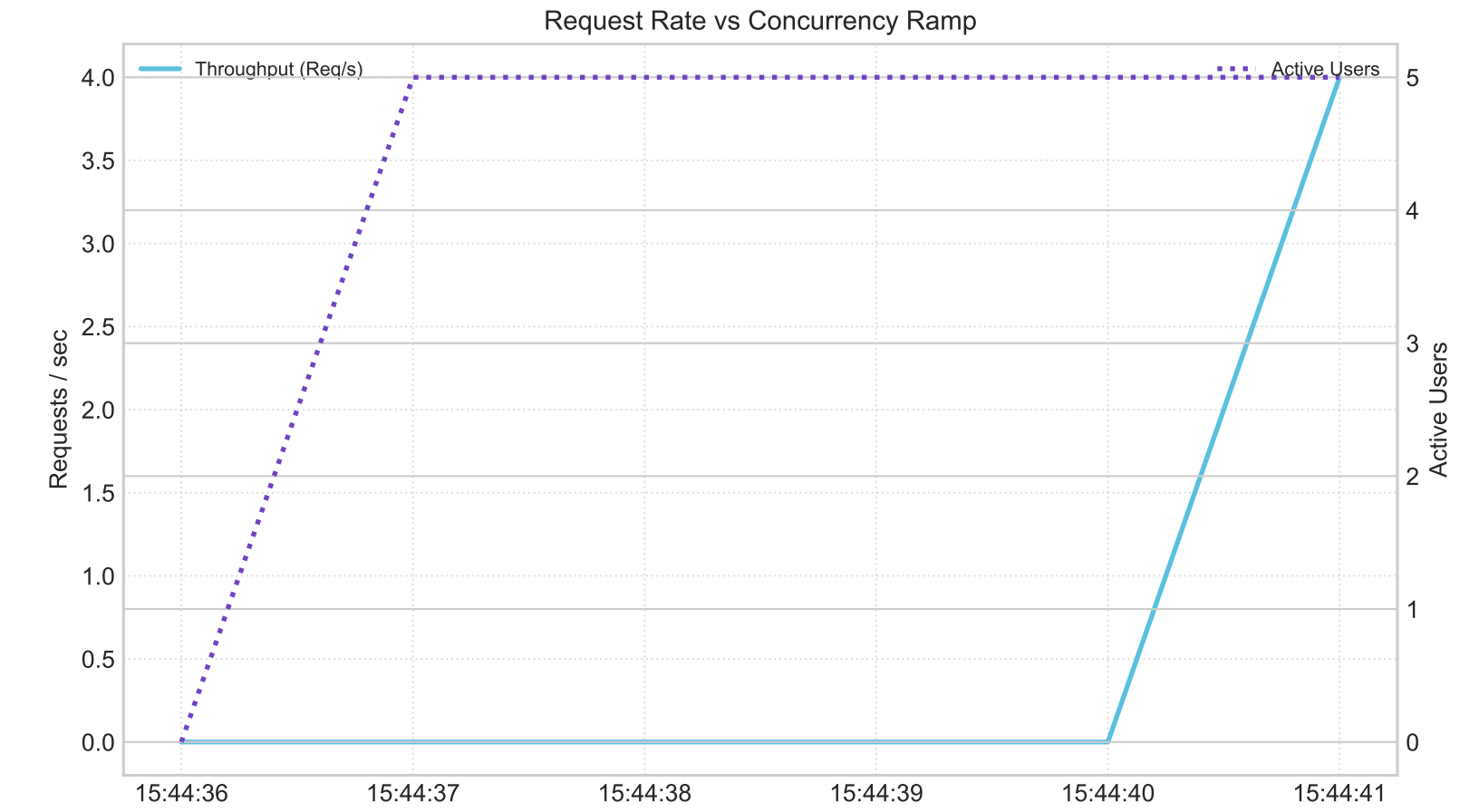
- Low request volume (41 requests); statistical latency metrics are tentative.
- P95 response time (1900.0 ms) violated SLA target (500.0 ms).
- Endpoint bottleneck identified on 'POST /login' with P95 of 1900.0 ms.

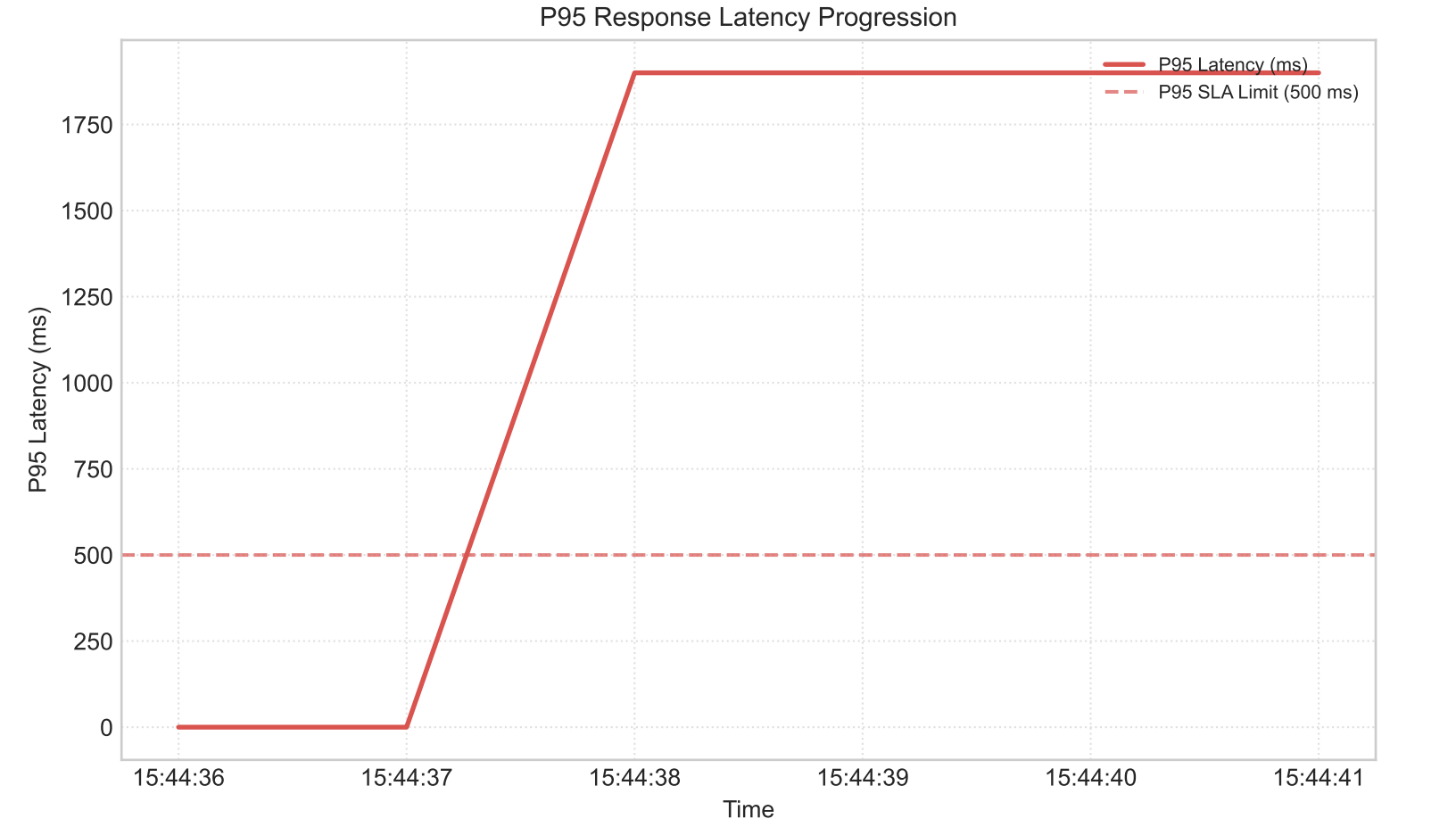
System & Hardware Resource Dynamics





Locust Workload & Latency Profile





Endpoint Breakdown & Chronological Timeline

HTTP Endpoint Performance Metrics:

Endpoint	Requests	Fails	Avg Latency	P95 Latency	RPS
GET /api/data	14	0	76.34 ms	140.0 ms	1.56
GET /health	9	0	10.55 ms	19.0 ms	1.0
POST /login	5	0	1885.21 ms	1900.0 ms	0.56
GET /me	12	0	13.19 ms	28.0 ms	1.34
POST /profile/update	1	0	61.69 ms	62.0 ms	0.11

Chronological Event Timeline:

Time	Event Type	Category	Severity	Description
15:[LOCUST] LOAD_STARTED	OBSERVED	Workload generation began with 0 users	INFO
[LOCUST] 15:[LOCUST]_PEAK_USERS	OBSERVED	Locust concurrency reached peak of 5 active u...	INFO
[SLA_ENGINE] 15:[SLA]_BREACH_DETECTED	INFERRED	SLA Target Breached: P95 Response Time was 19...	CRITICAL
[LOCUST] 15:[LOCUST]_PEAK_RPS	OBSERVED	Locust throughput reached peak of 4.0 req/s	INFO
[RUNNER] 15:[RUNNER]_EXPERIMENT_START	OBSERVED	Experiment started at (1787481523.1421537)	INFO
[PROMETHEUS] 15:[PROMETHEUS]_CPU_PEAK	OBSERVED	CPU utilization peaked at 0.54%	INFO
[PROMETHEUS] 15:[PROMETHEUS]_NETWORK_PEAK_TX	OBSERVED	Network transmit rate peaked at 0.0 MiB/s on ...	INFO
[PROMETHEUS] 15:[PROMETHEUS]_MEMORY_PEAK	OBSERVED	Memory utilization peaked at 16.77%	INFO
[PROMETHEUS] 15:[PROMETHEUS]_NETWORK_PEAK_RX	OBSERVED	Network receive rate peaked at 0.0 MiB/s on e...	INFO
[RUNNER] 15:[RUNNER]_EXPERIMENT_END	OBSERVED	Experiment completed at (1787481583.142155...	INFO